NexusRAG Enterprise Distributed System Architecture & SRE Standard

DEPARTMENT: ENGINEERING | VERIFIED ENTERPRISE POLICY

DOCUMENT SECTIONS (4 VERIFIED CLAUSES)

Section 1: Clause Index #1

System Architecture Overview: NexusRAG is architected as an asynchronous hybrid retrieval engine combining dense vector search via Qdrant, sparse BM25 indexing, and real-time reranking with sub-500ms latency SLAs.

Section 2: Clause Index #2

Service Level Objectives (SLOs): Production query availability SLA is 99.95%. Target p95 search latency is 450ms. Incident severity definitions are P0 (complete service outage), P1 (degraded retrieval or high error rates >5%), and P2 (non-critical pipeline delays).

Section 3: Clause Index #3

Security & Network Isolation: All inter-service communications enforce mTLS and mutual token verification. Multi-tenant department isolation is enforced cryptographically using RoleEnum claims in JWT payloads.

Section 4: Clause Index #4

Continuous Deployment: All service containers are built deterministically, scanned for CVEs via Trivy, and deployed progressively using canary deployments with automated rollback on healthcheck failure.